Multiple Mixer Based Multi Cycle Layer Composition for Power Efficient Display Pipeline

Shubham Pateria, Aseem Saxena, Krishna Kishore Jha
Multimedia and Systems Division
Samsung R&D Institute, Bangalore
Bangalore, India

Abstract— **In modern Graphic Architecture, display content is a mixture of different layers, analogous to a stack of cards which are mixed or composed pixel by pixel to form final display frame. Layer composition is handled by the GPU or an Overlay Hardware, either jointly or individually. While GPU can handle heavy composition, it results in higher power consumption as compared to Overlay. On the other hand, Overlay hardware is mostly constrained by the number of layers supported, causing dependence on GPU. Another issue with Overlay hardware is that many if its components are reserved for specific display operations such as external display support, and remain idle when such operations are not required. In this paper, we propose a technique which operates in multiple cycles to employ such under-utilized hardware components for the composition jobs. With this technique, the composition capability of mostly idle layer mixers can also be harnessed to handle more number of layers. The GPU load is thereby reduced resulting in less power consumption in the display pipeline while maintaining the normal display frame rate.**

Keywords— Display pipeline, Hardware Overlay, GPU, Mobile Display Processor, Graphics, Power Consumption

I. INTRODUCTION

In recent years, mobile computing devices have seen tremendous growth in terms of functionalities. Such devices are now packed with a multitude of user-targeted features and must support several applications. This expansion of functionalities demands more processing, continuous system operations and richer User Interfaces among other improvements. With such advancements, it becomes critical to address the issue of power consumption because mobile devices have to depend upon on-board power supplies.

Display pipeline plays a critical role in the aspects of system power efficiency and performance due to continuous processing involved in maintaining the rich user interface and power required by display peripherals [2], [3]. Therefore, optimization of display pipeline has been in focus of research and development, involving graphics rendering, windowing & composition, display processors, interfaces and panels, to ensure efficiency with as little power consumption as possible. Display content is typically a composition of different layers which are mixed according to their stacking order and transparency. The transparency property of a pixel may be implemented as an alpha-component which is an additional value to color information. The composition fundamentally involves alpha blending [11]. This scheme has become popular with the evolution of graphics architecture due to its benefits over rendering in a single shared buffer which requires more computational power for continuous windowing decisions made by the applications and more synchronization so that multiple applications do not override each other's display area [8]. The composition of various display layers can be performed by the GPU or an Overlay Hardware or both in a combined mode. Overlay hardware is a unit independent from GPU which is equipped with specialized hardware for mixing the layers. Usually, a Mobile Display Processor (MDP) is equipped with such Overlay unit along with other post-processing components [9]. GPU is packed with more processing power but MDP overlay mixer hardware is optimized for low-power composition jobs. Moreover, GPU based compositions require multiple fetch and write operations involving pixel buffers. The pixel processing on GPU cores and high bandwidth required for continuously moving the pixels contribute towards higher power consumption. On the other hand, MDP is mostly provided with composition support for only four layers to cover the basic UI of a mobile display – Status Bar, Launcher, Wallpaper and one application UI. This threshold is due to hardware and cost limitations on Overlay design. Thus, for many practical cases where UI is more layered, the system has to depend upon GPU for the composition, resulting in higher power numbers. Such issues demand for solutions which increase the efficiency of overlay hardware while ensuring minimal processing and bandwidth required by the GPU. In this paper, we have proposed one such method which performs layer mixing in multiple cycles or passes to increase layer threshold of the MDP hardware.

II. RELATED WORKS

Major issues related to composition with GPU arise from pixel processing load and bandwidth requirement for pixel data fetching and writing. This is covered in some detail in section VI.

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A common solution targeting the GPU load is Mixed Mode composition scheme. According to this method, composition can be performed in a shared manner where few layers can be batched and given to GPU and the rest are composed by MDP (for e.g. Video layer), thereby reducing the GPU processing load. Mostly, the layers with heavy pixel content are transferred to MDP.

Another technique optimizes the composition process for display content with low change rate by caching the composed frames [4]. In such Cache-based composition, if the display content remains unchanged for few cycles, the overlay layers are composed by the GPU and cached to be directly fetched and displayed in subsequent cycles. However, such optimization is only effective for static content and does not target all use cases in general.

Some prior proposals also optimize the composition process for Occlusion - targeting the fact that some visible features of overlay layers will be overlapping. For e.g. in a pop-up menu case, the menu will overlap the application background or wallpaper. Thus in most cases, number of input pixels from overlay layers is more than the number of output pixels in composed frame. Due to this, bandwidth requirement to fetch pixel data is redundant. A proposal by Khan et al. [1] suggests selective layer data fetching in case of full occlusion between layer regions by using descriptor arrays. In case of partial overlapping, fetching is done based on blending coefficients. Another similar proposal is for a display processor component tuned for selective pixel fetching using transparency based fetch masking [5]. One way to minimize bandwidth load is also to reduce the amount of reads and writes on a buffer by pre-determining pixel color data [7].Such techniques provide benefit of reduced bandwidth requirement by avoiding redundant occluded pixels.

Apart from system optimizations, separate hardware tuned for composition can also be provided. TI's OMAP4470 processor uses a separate Composition GPU (CGPU) whose hardware core is optimized for composition operations (like alpha blending) [12]. This reduces the load on main GPU unit and is also faster. Also, such hardware does not have typical limitations of MDP regarding layer threshold. However, this approach requires addition of extra hardware which contributes to cost.

The methods discussed above can reduce GPU load and bandwidth requirements. However, greater benefits can be obtained by relieving the GPU of composition. In other words, MDP should be capable of handling compositions itself. The main challenge here is the limited hardware capacity of MDP overlay. In this paper, we have proposed a composition technique which performs multiple cycles of layer mixing by employing the idle mixer hardware of MDP overlay unit. The proposed Multi Pass Composition (MPC) method enables MDP to handle higher number of layers while ensuring that the process does not affect the display frame rate (normally 60 FPS) or the system performance. This method is explained in section IV and V in details with comparison against usual technique. In section VII, experimental results for MPC have been discussed.

III. Display Pipeline Overview

Display cycle starts with various applications requesting graphics subsystem for their visual components or UI to be drawn. Applications (or activities, to be precise) can render display layers using different tools, such as skia or OpenGLES based hardware rendering (using GPU) [10]. The rendered data is saved into different graphic buffers assigned to various activities. These buffers can be moved around in the system using class objects called BufferQueues [8]. The layer buffers are managed by SurfaceFlinger (SF) service. Each activity connects with SF using BufferQueues as a producer of content. SF acts as a consumer for the graphic buffers. Once SF has the list of layers it asks the Hardware Composer (HWC) module to decide upon the composition. HWC is a component of Hardware Abstraction Layer (HAL). It acts as an interface between graphics framework and display kernel/hardware system. It decides whether the layers can be composed by overlay hardware in MDP. If such composition is not possible or partially possible, HWC marks all or some layers to be composed by GPU and notifies the SF service [FIG. 1]. Otherwise it prepares and manages the overlay HW for layer composition and display.

Different activities in the pipeline are tuned using a sync framework and occur over time periods called VSYNC for real-time efficiency and maintaining the required frame rate [8], [10]. For a 60 FPS display, VSYNC is around 1/60th of a second or 16.66 ms.

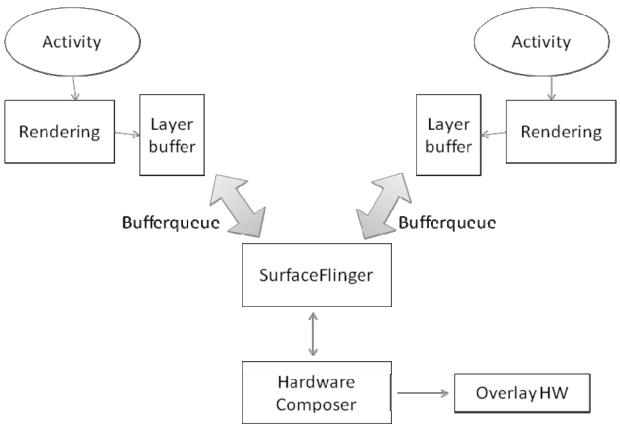


Fig. 1. Overview of Display Pipeline

IV. MDP Composition Architechture

The composition on Overlay HW is described here. A typical MDP Overlay is comprised of following components – i) hardware pipes which provide a guide or path for the flow of

image data towards display interface; ii) hardware mixers which perform composition upon the image layers passed through different pipes. The final display is thus obtained as a result of this composition.

A. Existing Composition Architecture

The current scheme of composition is described here for a general MDP overlay unit consisting of four pipes and two mixer components in the hardware.

The four pipes are marked here as PIPE1 to 4. Four display image layers queued to the pipes are marked as A to D [Fig. 2]. These are the layers received from HWC as per process described in section III [Fig. 1]. Two layer mixers are typically used – LM0 used for the primary display interface and LM1 connected to a Writeback path which saves composed data in an output buffer for secondary display like Wi-Fi Display (WFD). In usual operation where no secondary display is active, LM1 remains idle.

After HWC identifies that MDP composition is possible, the layers A, B, C and D are queued to PIPE 1 to 4 as shown in Fig. 2. This happens while a previous composed frame is being displayed. After queuing, the process will wait for a VSYNC event to arrive before layer mixing begins. Once this event is received, the queued layers are passed to LM0 which mixes the data of separate layers into a single composed frame. This frame is passed through primary interface to the display. The mixing and display operation also lasts for one VSYNC period. Thus, in two sync periods, the MDP composes four layers and displays a single frame.

This scheme involves a single pass of display layers and thus the number of layers supported is constrained by hardware limitation on the number of pipes and mixer capacity. In the above case, MDP cannot support composition for more than four layers.

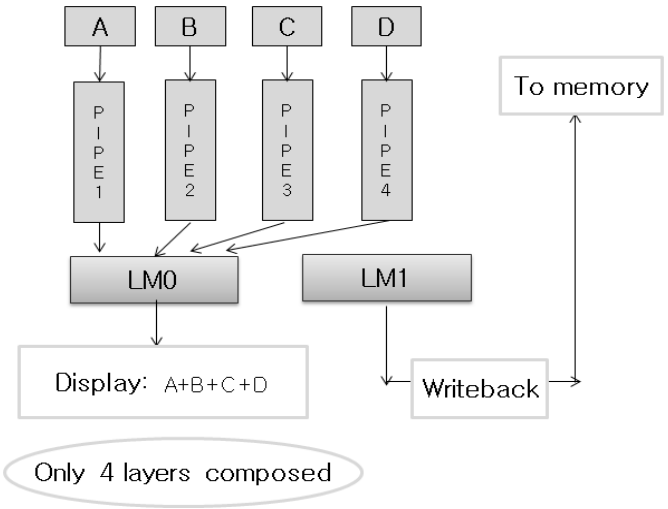


Fig. 2. MDP Overlay Composition Process for four layers

B. Proposed Composition Architecture

The proposed method of Multiple Mixer based Composition or Multi Pass Composition (MPC) performs layers composition in multiple passes as opposed to the single pass in existing scheme. This is achieved by employing LM1 for the composition job along with LM0 when WFD or any other secondary display is not in use.

The multiple cycles of composition have been described here. Pipes and mixers are marked as above. Additionally, a buffer bank is used for storing LM1 compositions, marked as X, Y, Z. Initially, 3 layers are queued to the hardware pipes while one remaining pipe is used to transfer previously composed frame (X) [Fig. 3].

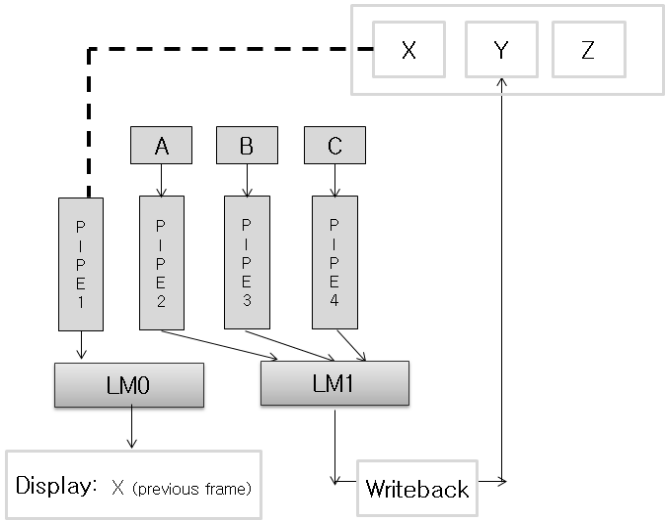


Fig. 3. First pass of MPC

The queued layers are composed by the mixer LM1 into a first frame. This frame is passed through the Writeback and saved into buffer Y. In the second cycle (or pass), the first frame Y is queued along with 2 other layers – D & E- to the pipes [Fig. 4]. Again, these layers are fetched by pipes and composed by LM1 into a final frame which is now a composition of layers A, B, C, D and E and stored in buffer Z. The above two passes are completed within one vsync period.

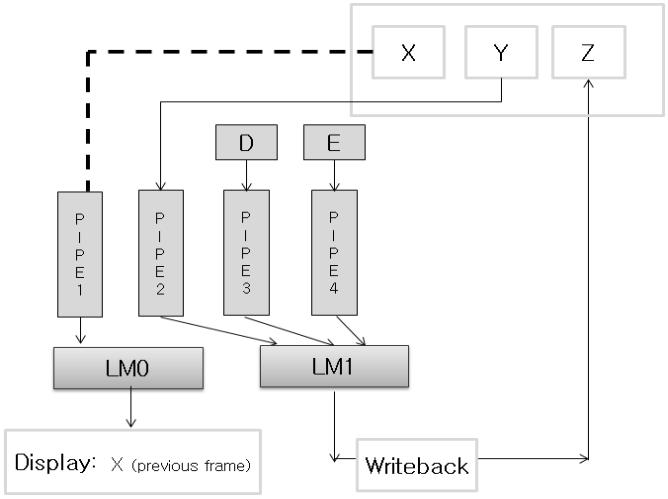


Fig. 4. Second pass of MPC

In the next vsync period, buffer Z is queued to PIPE0 and composed frame is fetched to be displayed on the primary panel. [Fig. 5]

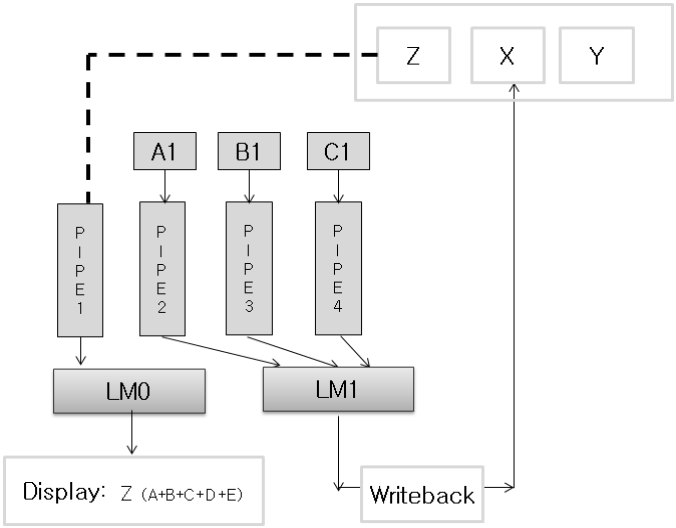


Fig. 5. Final Display of Composed frame

As described, in multiple passes, the MPC process is able to compose five layers as opposed to four layers by existing technique within two vsync periods from queueing. Thus, higher layer threshold can be achieved for MDP hardware while maintaining the required frame rate.

V. Software Process

The operations of the MDP overlay hardware are controlled by a decision and configuration process handled by Hardware Composer (HWC). In this section, the process is briefly described along with our proposed algorithm.

A. Existing Composition Process

The usual flow of logic is shown in Fig. 6 for MDP Overlay control. After SurfaceFlinger passes new layers, the HWC must decide whether composition can be handled by MDP. The layer count limit of MDP is crucial for this decision. This decision is made in the Prepare phase. If MDP composition is not found possible, the layers are immediately redirected towards SF flagged with the condition that GPU composition is required. On the other hand, if MDP composition is possible, then in the prepare phase the MDP pipe configurations are decided according to layer content. In the Set phase, the layers are queued to the assigned pipes. Each hardware pipe saves the buffer address of respective display layer from which image data will be fetched. Once the process of configuration and queuing is finished, the final commit call is made on arrival of VSYNC event to send data for composition and display.

In case of full GPU composition, the overlay operations are only performed for one composed frame received from GPU. Layer mixer does minimum job (mainly fetching) in this case.

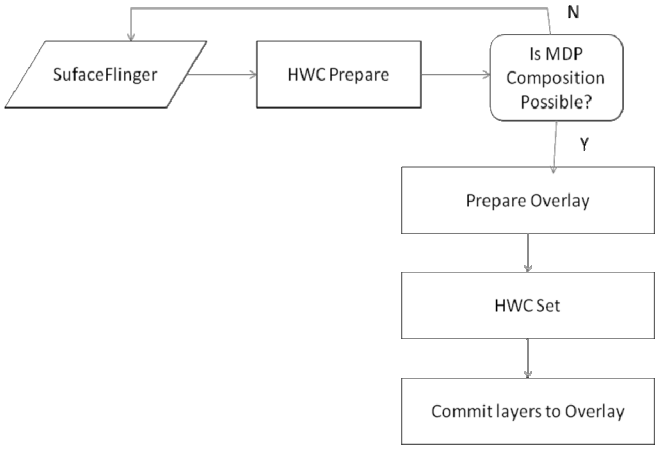


Fig. 6. HWC process flow for normal composition

B. Proposed MPC Process

In the proposed MPC process architecture, the Prepare and Set phases of HWC are further divided to handle multiple passes (Fig. 7). As HWC Prepare begins, the validation for MDP composition feasibility is modified according to the new layer count threshold after implementation of MDP. Moreover, WFD should not be in use for MPC to work so that Writeback path and mixer are free. If these conditions are met, the overlay and writeback interface are initialized. First batch of layers are taken (A, B, C in Fig. 3) and pipe configurations are decided accordingly.

In Set phase, the layers in the first batch are queued to respective pipes which are configured to use LM1. An output buffer is queued to the writeback path. This acts as the storage for data passed through writeback. In order to support multiple passes, more than one output buffer would be required. In our

test case, at least two buffers are created to support two writeback passes. Apart from this, one more buffer is created to hold the final frame to be displayed on the primary panel. After queuing is done, commit call is made to pass the layers through the pipes to LM1 and write-back the composed frame to the output buffer.

Once the contents are written to output buffer, it is dequeued from writeback path. This finishes the first pass. A new batch is then taken for the second pass containing the remaining display layers and the composed frame from the first pass. The queue-commit-dequeue process is repeated for this batch (Pass 2). The composed frame is written to new output buffer. At the final pass, there is one output buffer containing a composed frame of five layers from two passes. In order to display the contents on the primary panel, one overlay pipe is configured according to final frame's configurations. After final commit, the composed frame is fetched by LM0 and displayed on primary panel.

Whether in normal mode of operation or while using the MPC scheme, the Prepare and Set phases must finish within one VSYNC period in order to maintain proper frame rate. In MPC, each of the two writeback passes before final commit has been found to finish in 5-6 ms avg. Thus, all the passes finish within the allotted time period of a VSYNC. Time required by different process steps is shown in Table IV.

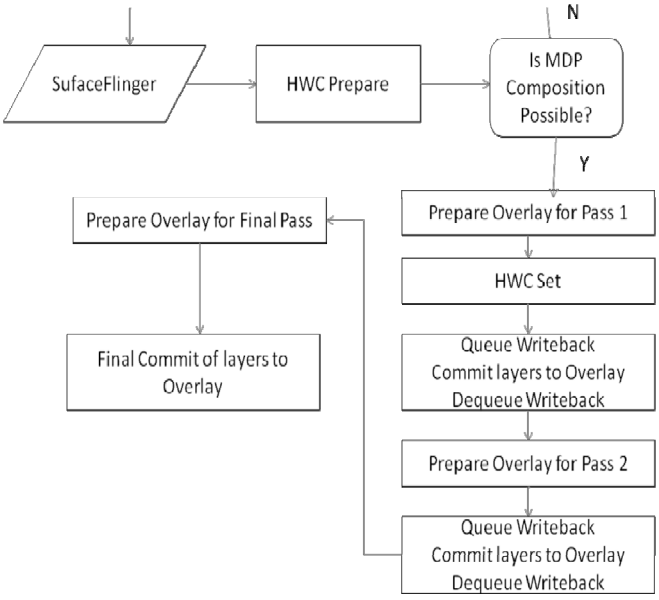


Fig. 7. HWC process flow for MPC based composition

C. MPC selection and tuning logic

The decision process to shift to MPC based composition takes few important factors into consideration, apart from the layer count. Following design structure discusses the decision logic for MPC for the five layer example discussed above. This is for a display processor with four pipes – 2 RGB, 1 DMA (direct memory access) and 1 VIG. Only VIG pipe is capable of scaling the layers.

Basically, apart from WFD and layer count constraints, the pipe support needs to be considered. Using two passes, MPC can compose two VIG format layers (like video or live wallpaper). However, their z-orders must be as one in range 1 to 3 and other in 4 to 5. This is to ensure that two VIG layers are not passes together because only one VIG pipe is available. Same applies to layers requiring scaling. These constraints can change according to MDP hardware properties and need to be modified accordingly.

If MPC is found feasible, the MDP clock frequency is boosted to push layers in multiple passes. This has minimal effect on power consumption.

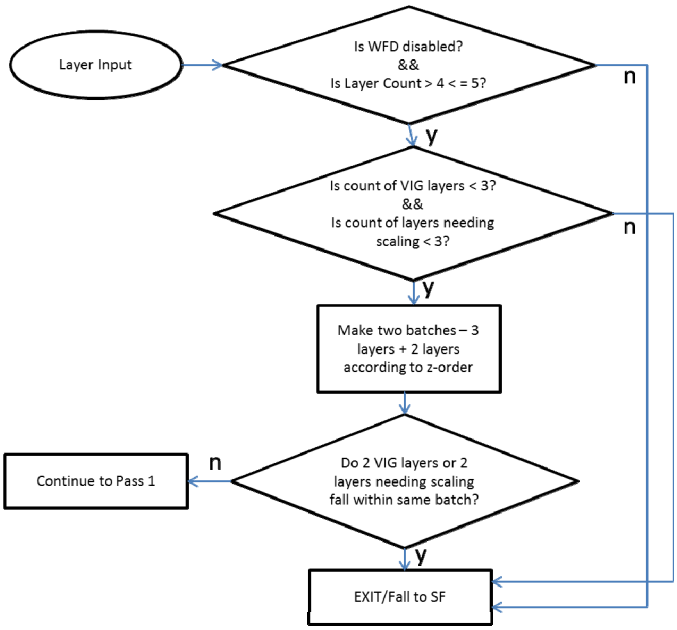


Fig. 8. MPC selection logic

Using the process discussed, MPC is able to achieve MDP overlay based composition for more layers compared to traditional method.

VI. GPU BASED VS MDP BASED COMPOSITION

There is a couple of major problems with using GPU as a dedicated resource for composition. Firstly, mobile GPU is designed to handle rendering operations and computation on primitives. When it is simultaneously used for composition, the computing load on GPU may reach near maximum because due to excess processing over heavy rendering for the UI. Unless hardware or process optimizations are performed, this would often lead to high power consumption. As discussed earlier, TI OMAP employed extra CGPU unit to provide optimized hardware for composition specific jobs. However, MDP is another hardware whose mixers are

optimized for composition and because all devices have display hardware interface, MDP is widely available (as compared to CGPU).

Secondly, GPU based composition requires multiple memory read/write operations which require bandwidth. Table I shows the bandwidth requirements for a standard Home Screen case for a resolution of 720x960. Bus bandwidth is mainly consumed in three layer data fetches and writing composed frame to framebuffer, all these operations being simultaneous. At the same time, frame data from previous framebuffer is being fetched by MDP. Layer buffer read is a general operation but framebuffer write comes as an overhead. MDP solves this problem by using layer mixers which lie in the path of the layer data moving to display interface. By doing composition on-the-fly, excess read/writes are avoided. The only major bandwidth utilized in this case is that required for layer data fetch.

TABLE I. HOME SCREEN COMPOSITION – FULL GPU MODE

Operation	*Calculation*	*Bandwidth (MBps)*
Wallpaper Read	720x960 x4(bytes/pixel) x 60 fps	170
UI Launcher Read	720x960 x4(bytes/pixel) x 60 fps	170
System Bar Read	720x45 x4(bytes/pixel) x 60 fps	8
Composited output write to FB	720x960 x4(bytes/pixel) x 60 fps	170
FB scan-out to display	720x960 x4(bytes/pixel) x 60 fps	170
	Total	688

Table II shows bandwidth requirements for Home Screen case when using MDP based composition.

TABLE II. HOME SCREEN COMPOSITION – MDP OVERLAY

Operation	*Calculation*	*Bandwidth (MBps)*
Wallpaper Read	720x960 x4(bytes/pixel) x 60 fps	170
UI Launcher Read	720x960 x4(bytes/pixel) x 60 fps	170
System Bar Read	720x45 x4(bytes/pixel) x 60 fps	8
	Total	348

Thus, significant bandwidth is saved (nearly 50% for above case) when using MDP overlays for composition. However, this is not the only benefit of using MDP. The MDP layer mixers are optimized hardware units designed for power efficient composition. Offloading composition to these mixers leaves GPU for only rendering which means reduced GPU load and less power consumed in layer blending. But, the caveat here is the limited layer capacity of MDP, as discussed earlier.

In order to test the power benefit of MDP overlay against GPU composition, comparative experiment was performed for up to four layer composition. Display resolution used is 720x960. Setup includes device equipped with Qualcomm MDP having four pipes and one primary mixer compared against Adreno GPU. The results are provided in Table III.

TABLE III. MDP VS GPU POWER CONSUMPTION

No. of Layers	Average Current (mA)	
	MDP based Composition	*GPU based Composition*
1	384.76	400.64
2	390.96	410.59
3	437.68	443.86
4	459.96	481.58

This data has been visualized in the line chart shown in Fig. 9.

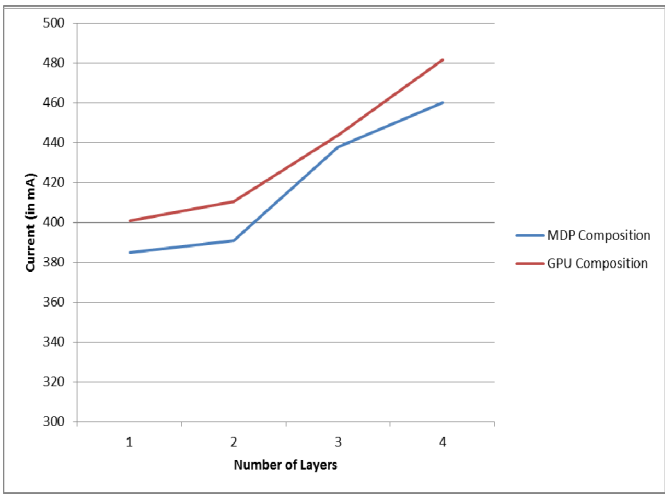


Fig. 9. MDP vs GPU Power Comparison

Increasing the number of layers would effectively increase the bandwidth requirement (amount of data) and computations (number of pixels). This means higher power consumption. However, while GPU and MDP powers increase with the number of layers, it can be seen that MDP always consumes less as compared to GPU.

VII. VALIDATION AND EXPERIMENTAL RESULTS

The results discussed in this section were obtained using following setup – Qualcomm MSM8916 chipset equipped devices with Qualcomm MDP and Adreno GPU. The display resolution was 720x960. The power measurements were performed for Animation app rendering five layers at a time.

A. Multipass Composition Timings

In a usual process, two vsync periods are consumed before layers submitted by the Surfaceflinger are displayed on the screen as a single frame. First, the SF and HAL (mainly HWC) operations prepare the display processor to handle upcoming

layers and in the subsequent vsync period, the hardware fetches layer data, composes and displays. [Fig. 10]

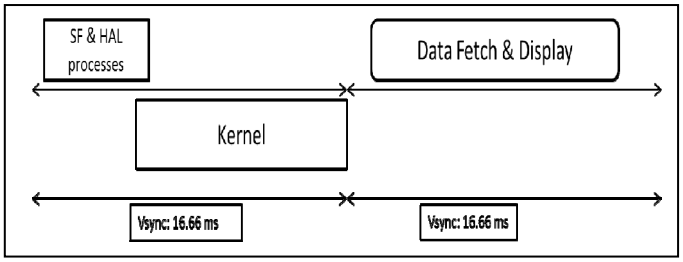


Fig. 10. Processing and fetch/display stages for one set of layers – Normal mode

The SF and HAL operations run on *surfaceflinger* thread. Main operations are MessageReceive, HWC Prepare, Set, merge etc. The operations of our focus are *Prepare and Set*. Henceforth, operations apart from these will be referred to as *other operations*. These operations are performed on each vsync cycle. However, it can be seen from Fig. 11, that there is a gap between two consecutive processes. This gap may vary according to complexity of the different process stages. MPC exploits this gap period to fit the HWC operations necessary for multiple passes. Data passes through writeback interface can be forced unlike vsync-controlled fetch performed by primary interface. Thus, we are able to perform multiple passes through writeback as extension of the HWC process itself, instead of waiting for next vsync to initiate data fetch. [Fig. 12]

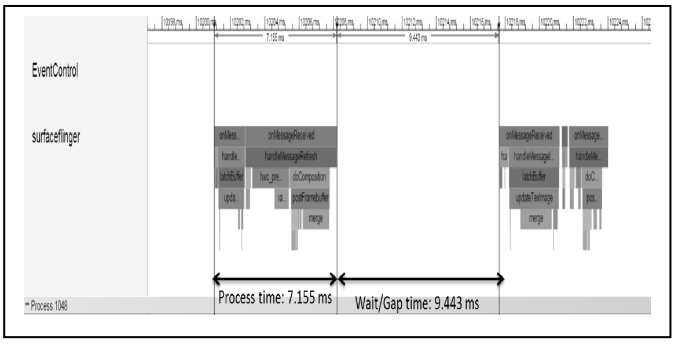


Fig. 11. Processes and wait period on *surfaceflinger* – Normal mode

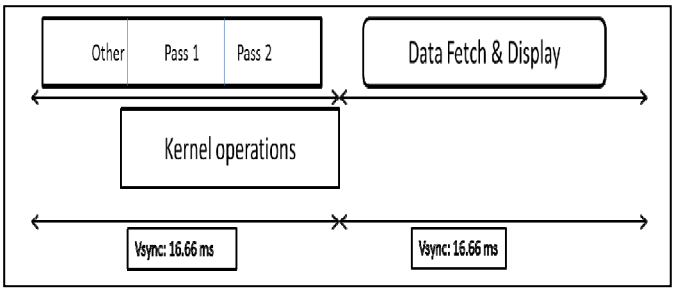


Fig. 12. Processing and fetch/display stages for one set of layers – using MPC

TABLE IV. TIME PERIODS FOR MAJOR OPERATIONS BEFORE DISPLAY

Mode	*Other operations (ms)*	*Prepare phase (ms)*	*Set Phase (ms)*		*Wait (ms)*	*Total (ms.)*
Normal Mode	5.97	0.483	1.948		10.69	16.66
MPC with custom animation application	6.42	0.578	13.834		0.15	16.66
			Pass 1	Pass 2		
			7.94	6.60		
MPC with OpenGL Benchmark	5.26	0.237	13.97		1.40	16.66
			Pass 1	Pass 2		
			5.87	7.21		

Table-IV shows a comparison between time required for different operations in normal mode and while using MPC. The MPC timings were recorded for a five-layer custom animation app and Open GL Benchmark app. It can be seen that all passes of MPC still finish within the required VSYNC time (16.66ms).

B. MPC vs GPU based Composition

Comparative data for 5 layer composition experiment is depicted in Fig. 13. The power measurements were performed in three modes – full GPU based, mixed mode (common techniques) and using MPC. Rendering is hardware accelerated which means GPU does the major computation for drawing the five UI layers. If Full GPU composition is used, there is overhead of computation (pixel blending) and bandwidth. Shifting to MPC does not necessarily reduce bandwidth. Read operations from writeback memory can still be overheads. The main idea is to utilize layer mixers (which consume less power) for composition instead of GPU. Mixed Mode does a nearly similar job. Unless the layers sent to GPU are significantly small (like status bar) and overlapping, bandwidth saving is small in Mixed Mode against GPU bandwidth because FB write is still needed. In such cases, power savings in Mixed Mode are due to power efficient layer mixers. However, compared to MPC, Mixed Mode still uses GPU's computing resources. Due to these reasons, MPC saves power compared to both full GPU and Mixed Mode compositions.

It can be seen from Fig. 13 that for a five layer composition, MPC saves nearly 50 mA against GPU and 20 mA against Mixed Mode. Thus, MPC is more power efficient as expected.

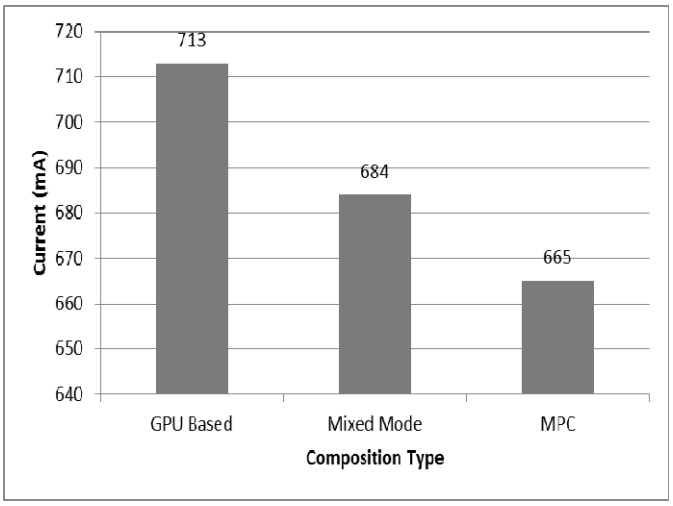


Fig. 13. Comparison of the current requirements for different composition schemes. Current values are in mA.

The optimization achieved by MPC is not just limited to four-piped display processors. By applying this technique to advanced processors with eight or ten pipes, we can achieve significant composition improvements in such processors as well. The layer thresholds achievable for different Qualcomm chipsets are provided in Table V.

TABLE V. MPC LAYER SUPPORT FOR DIFFERENT PIPE COUNTS

Chipset	Resolution	Pipe Count	Current layer support (based on mixer capacity)	Layer support with MPC
8916	720x1280	4	4	5
8939	1080x1920	4	4	5
8974	1080x1920	8	4	7
8994	1080x1920	10	8	14

VIII. CONCLUSION

In this paper, we proposed a Multi Pass Composition (MPC) scheme which uses the idle layer mixer and writeback components of Mobile Display Processor (MDP) hardware in order to increase the number of overlay layers handled by MDP for composition. By increasing the layer threshold, MDP can handle composition for complex UI and the involvement of GPU is reduced. Through experiments, it was found that MPC technique is more effective as compared to full or partial (Mixed Mode) GPU based composition. For the five layer composition test cases, MPC achieved about 40-50 mA reduction in power consumption against full GPU composition and about 20 mA reduction against Mixed Mode composition. Various stages of MPC process were found to complete within standard sync periods and thus display frame rate is not affected. This optimization is significant considering that UI is not always simplistic and many use cases require five-layered display such as menu pop-up, video pop-up etc. Reducing GPU involvement would be beneficial in such cases. This method is expandable to advanced MDP hardware equipped with more pipes and mixers enabling more than five layers and Multi-window support using Overlay hardware and limited GPU usage. The method can also be clubbed with other existing optimization schemes such as display frame caching and mixed mode composition to achieve more efficiency in the display pipeline.

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